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Cartão: 00326048

Tarefa 5

Fs (Hz)	Freq sinal original (Hz)	Freq do ruído (Hz)	MSE
600K	*3K*	*120K*	*2.000989e-02*
300K	*3K*	*120K*	*2.001978e-02*
100K	*3K*	*20,1K*	*2.004821e-02*
48K	*3K*	*?*	*4.000000e-02*
16K	*3K*	*?*	*4.000000e-02*
9,6K	*3K*	*?*	*4.000000e-02*
2K	*900*	*?*	*4.000000e-02*

Observação: A partir de Fs = 100KHz podemos ver que a frequência do ruído começa a ser rebatida para esquerda, isto pelo fato da frequência de amostragem não respeitar pelo menos 2x a maior frequência do sinal, no caso o ruído. No último caso, Fs = 2KHz, podemos ver o mesmo para o sinal original.

Fs = 600KHz

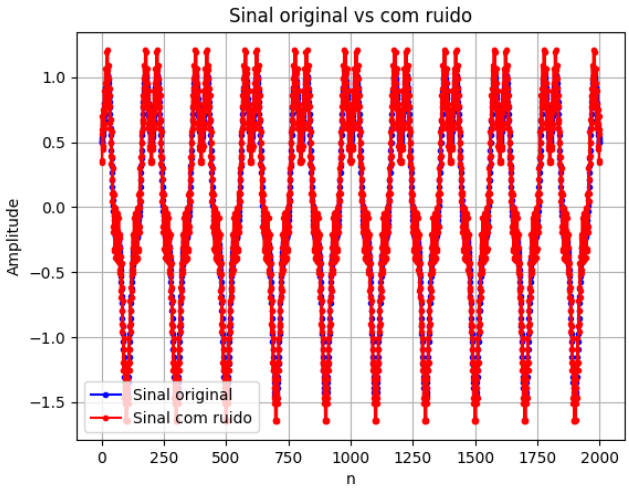

Figura 1 – Sinal original vs com ruído

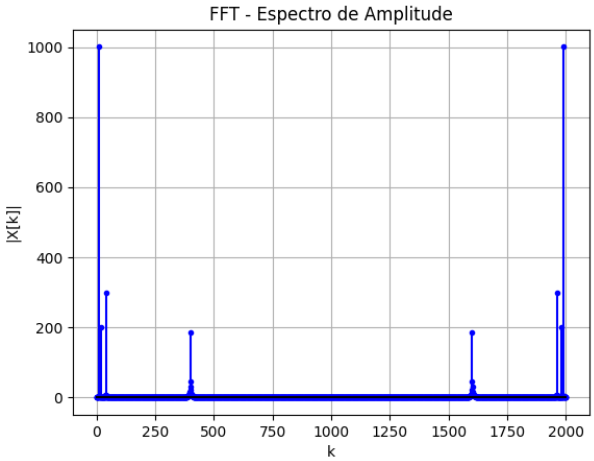

Figura 2 – FFT do sinal

Fs = 300KHz

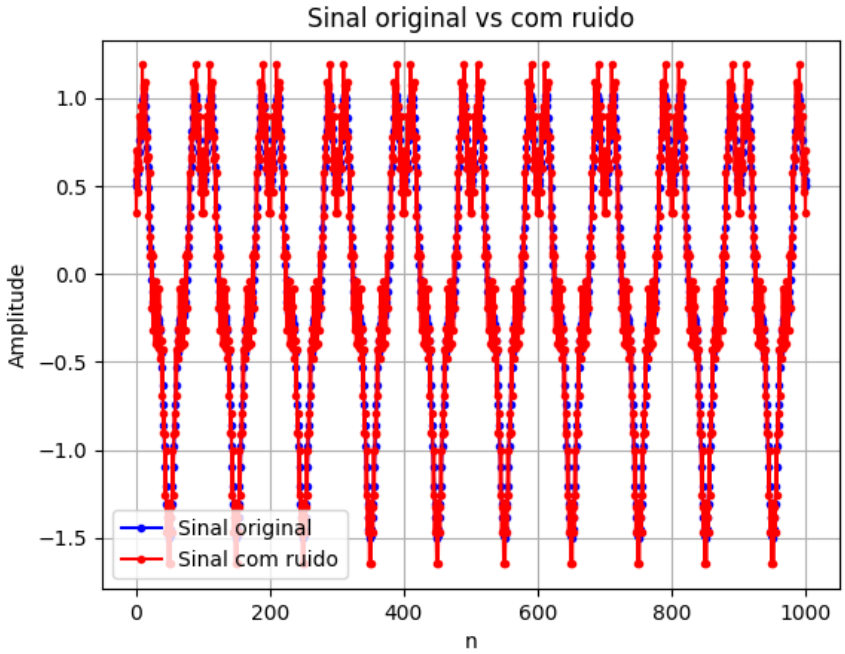

Figura 3 – Sinal original vs com ruído

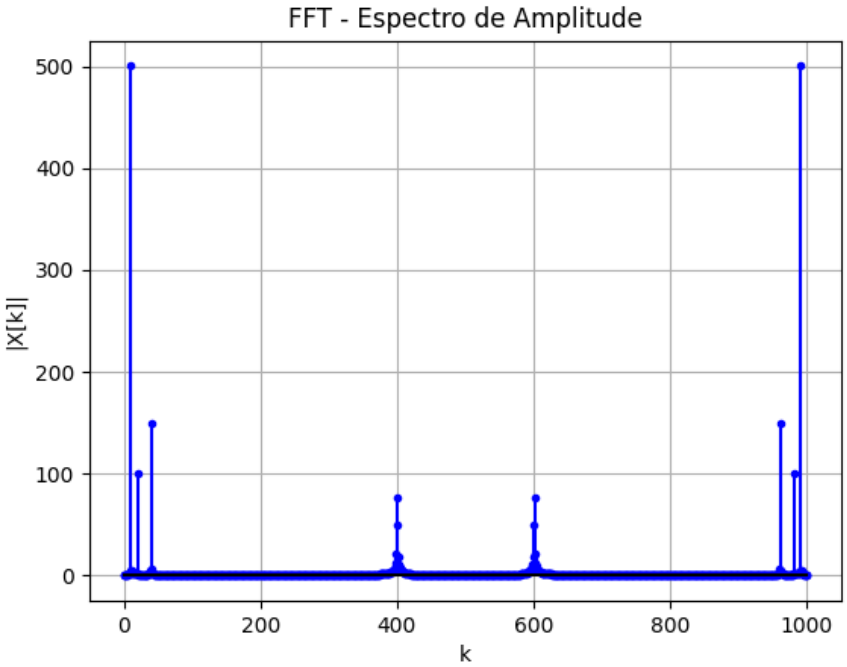

Figura 4 – FFT do sinal

Fs = 100KHz

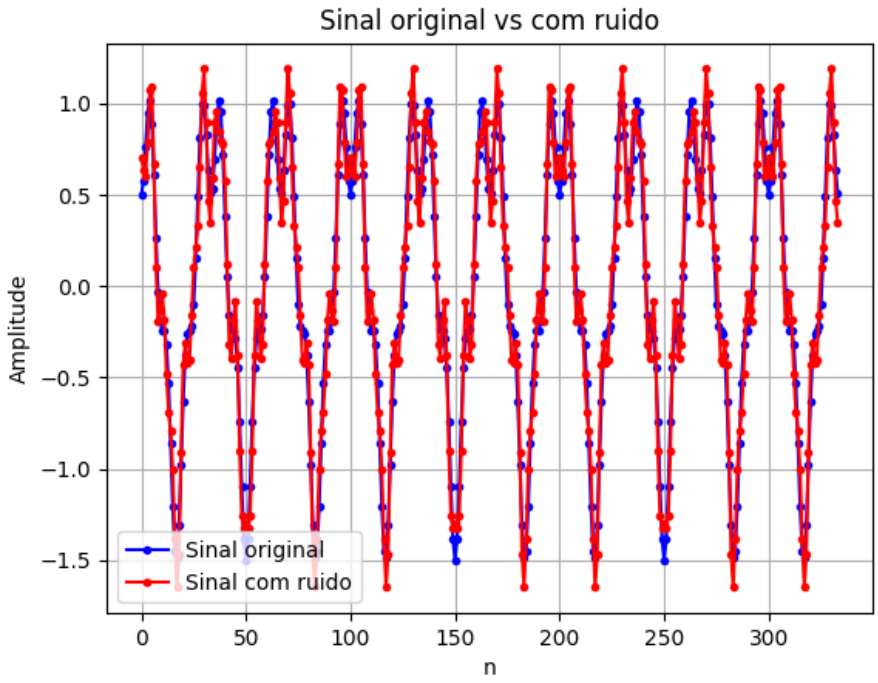

Figura 5 – Sinal original vs com ruído

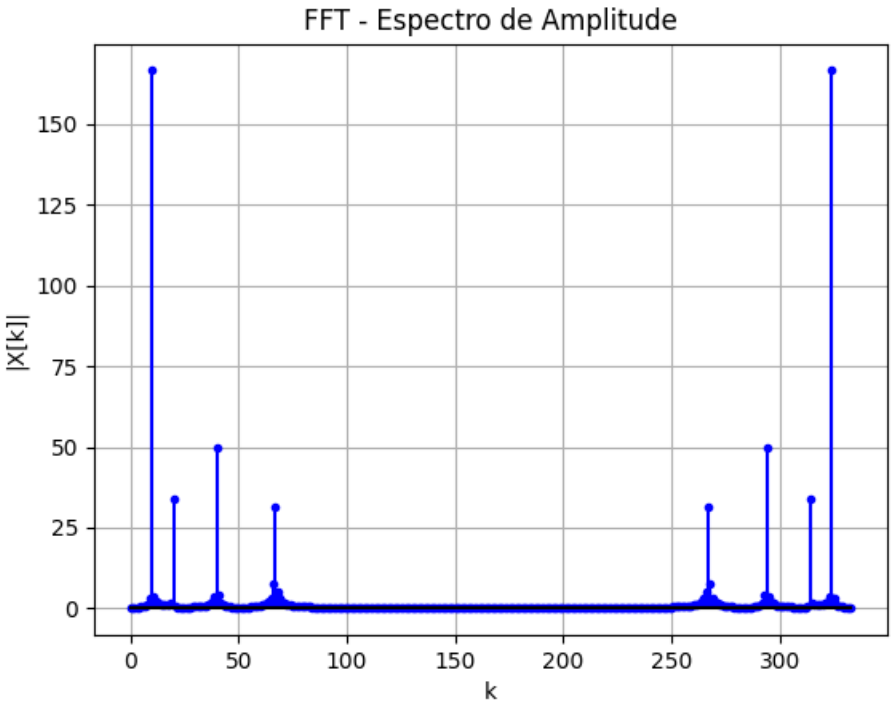

Figura 6 – FFT do sinal

Fs = 48KHz

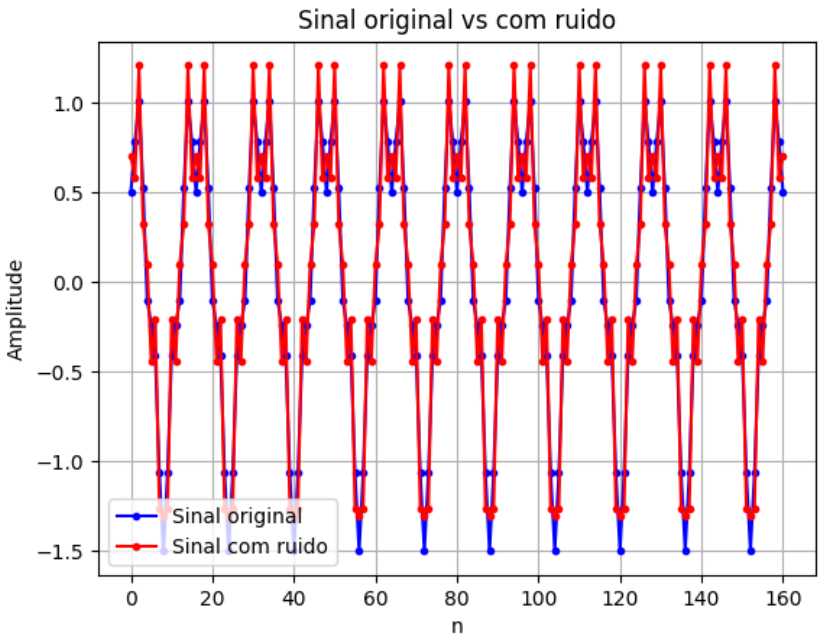

Figura 7 – Sinal original vs com ruído

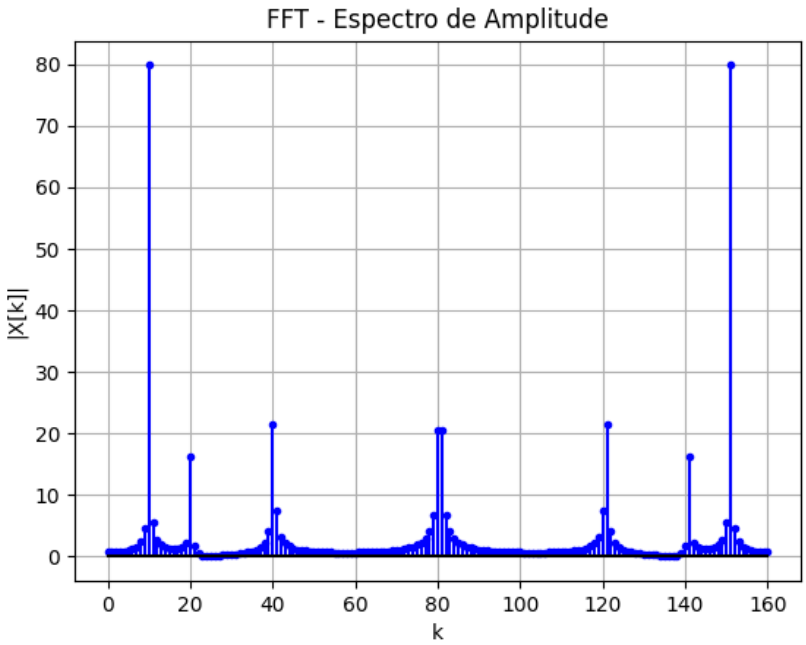

Figura 8 – FFT do sinal

Fs = 16KHz

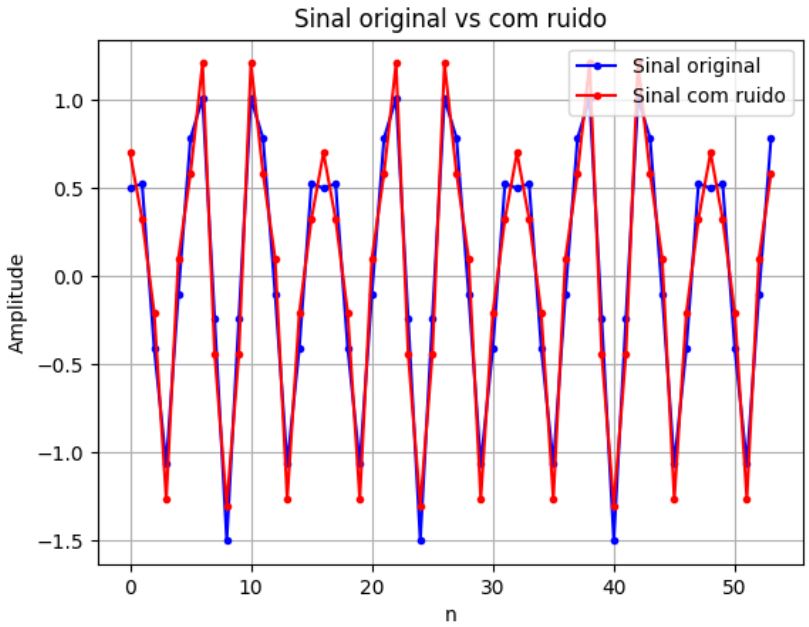

Figura 9 – Sinal original vs com ruído

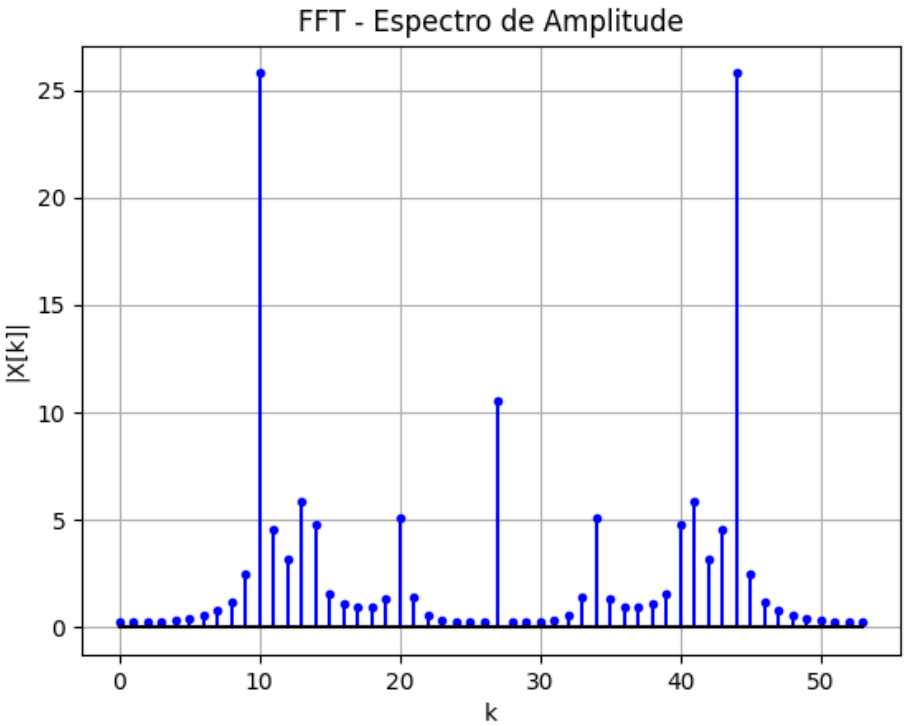

Figura 10 – FFT do sinal

Fs = 9,6KHz

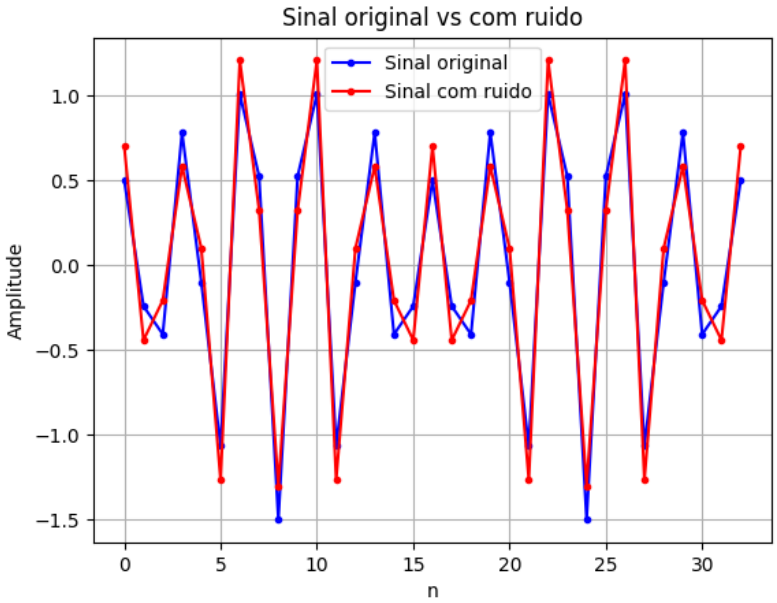

Figura 11 – Sinal original vs com ruído

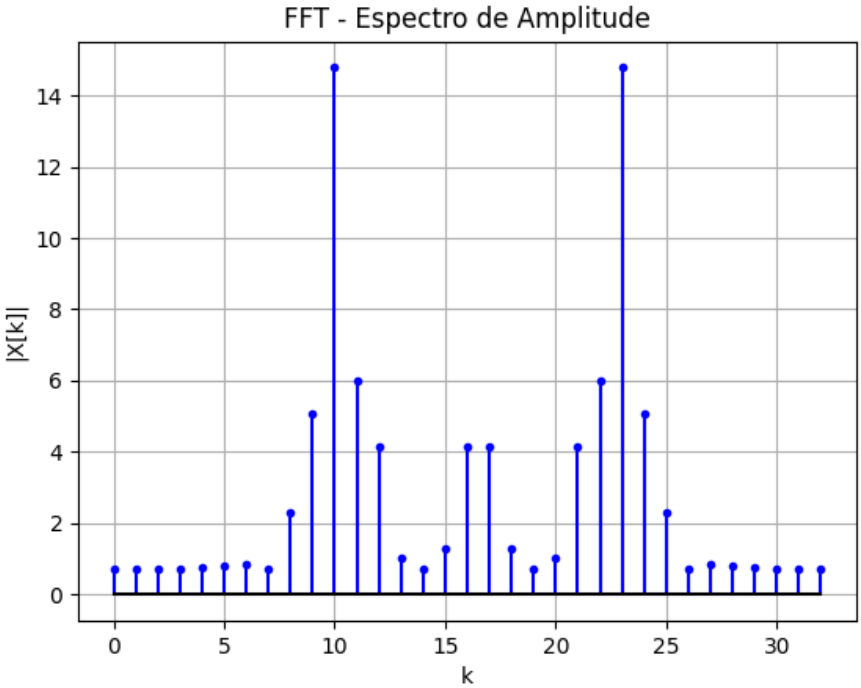

Figura 12 – FFT do sinal

Fs = 2KHz

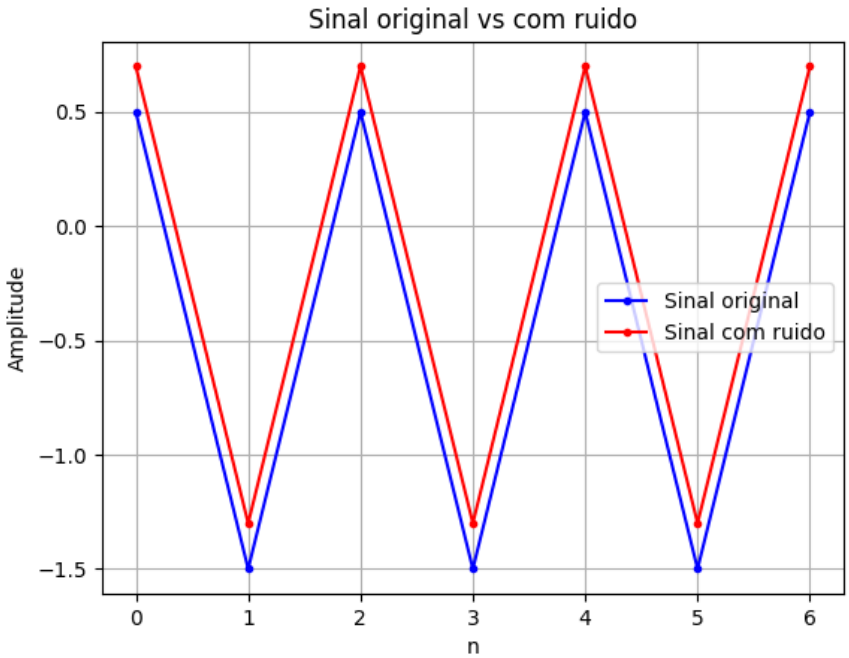

Figura 13 – Sinal original vs com ruído

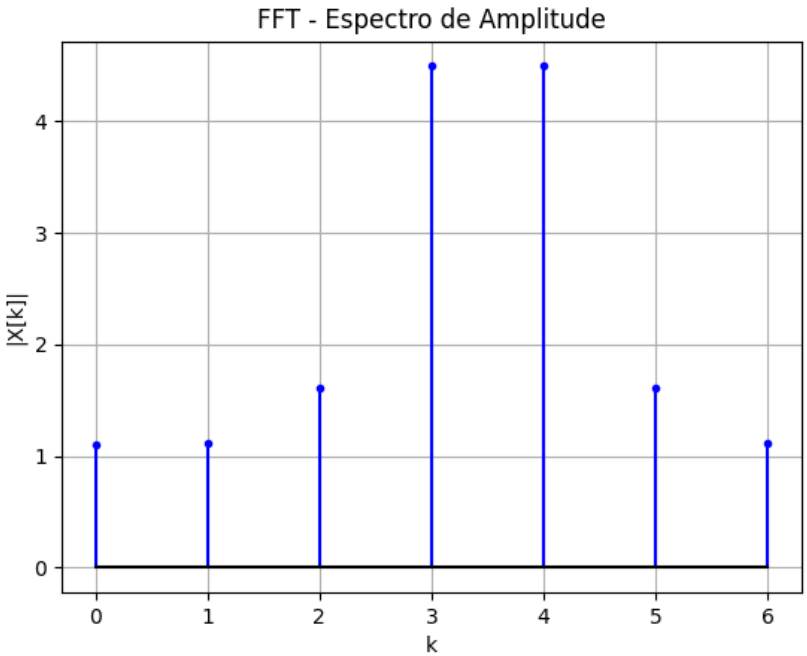

Figura 14 – FFT do sinal